

Linear Modeling & Residuals

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **calculate and interpret** average rate of change with appropriate units.

RELATED SKILLS

- ☐ D Calculate average rate of change over an interval.
- ☐ D Interpret average rate of change with contextual units.
- ☐ A Attach and interpret units for rates, inputs, and outputs.

- ☐ 2. I can **construct** a linear function from two data points, a table, a graph, or a context.

RELATED SKILLS

- ☐ A Calculate slope from two points.
- ☐ A Explain a model parameter in the language of its context.
- ☐ R Convert among common forms of a linear equation.

- ☐ 3. I can **fit** a linear regression model to data using technology and interpret its parameters.

RELATED SKILLS

- ☐ I Use technology to determine a least-squares linear regression model.
- ☐ D Explain a model parameter in the language of its context.

- ☐ 4. I can **interpret** a correlation coefficient as evidence about the direction and strength of a linear relationship.

RELATED SKILLS

- ☐ I Interpret the sign and magnitude of a correlation coefficient in context.

- ☐ 5. I can **calculate** residuals, construct a residual plot, and use its pattern to evaluate a model.

RELATED SKILLS

- ☐ I Calculate an observed value minus its model-predicted value.
- ☐ I Construct and interpret a residual plot.
- ☐ I Use residual patterns and contextual evidence to judge whether a model is appropriate.

- ☐ 6. I can **use** a linear model to make and interpret predictions, including solving for either variable.

RELATED SKILLS

- ☐ D Solve for an input associated with a specified output.
- ☐ A Identify input and output values in a formula, graph, table, or context.
- ☐ A Restrict a model to meaningful contextual inputs.